Problems marked with a (*) are a bit more complex.

7.1. Diffeomorphism.

(a) Give a diffeomorphism between $\mathbb{R}^2$ and $(0,1) \times (0,1)$.

(b) Is $f(x) = x^5, x \in \mathbb{R}$ a diffeomorphism of $\mathbb{R}$ to itself? Motivate rigorously your answer.

7.2. Inverse function I. Consider the function $F : \mathbb{R}^2 \to \mathbb{R}^2$ given by $F(x,y) = (x^2y, xy^2)$. Show that F is locally invertible around all points (x,y) such that $x \neq 0$ and $y \neq 0$. Compute the differential of the local inverse of F at the point $F(2,1)$.

7.3. Implicit Function I. Sketch the zero set of the following functions $f : \mathbb{R}^2 \to \mathbb{R}$:

1. $f(x,y) = x^2 + y^2 - 1$,
2. $f(x,y) = y^2(1-x) - x^3$,
3. $f(x,y) = y^2 - x^2(x+1)$,
4. $f(x,y) = xy(x+y-1)$,
5. $f(x,y) = x^2y^2 - x^2 - y^2 + 1$.

You can also use software to help you.

At which points $(x_0, y_0) \in \mathbb{R}^2$, does the Implicit Function Theorem imply that the function can be locally resolved with respect to x (or with respect to y, with respect to both, or possibly with respect to neither variable)? Mark these points in your sketch.

7.4. Multiple choice. Mark all and only the true statements

(a) Let $U \subset \mathbb{R}^n$ be open and $f \in C^1(U, \mathbb{R}^n)$ such that $\det Jf(x) > 0$ for all $x \in U$. Then the set $f(U)$ is open.

(b) Let $U \subset \mathbb{R}^n$ be open and $f \in C^1(U, \mathbb{R}^n)$ such that $\det Jf(x) > 0$ for all $x \in U$. Then f is injective.

(c) Is there a diffeomorphism $\phi\colon \mathbb{R}^2 \to \mathbb{R}^2$ such that $\phi(U) = V$, with $U := \{x^2 + y^2 < 1\} \subset \mathbb{R}^2$ and $V := \{x^2 + y^2 \leq 1\} \subset \mathbb{R}^2$.

(d) (*) Let T be a triangle and Q be a square on the plane (just the boundary, not the interior). Is there a diffeomorphism $\phi\colon \mathbb{R}^2 \to \mathbb{R}^2$ such that $\phi(T) = Q$?

7.5. Implicit function II. Show that the system of equations

$$\begin{cases} xy^5 + yu^5 + zv^5 = 1, \\ x^5y + y^5u + z^5v = 1, \end{cases}$$

is solvable for the variables u and v in a neighborhood of the point $(x_0, y_0, z_0, u_0, v_0) = (0, 1, 1, 1, 0)$ and determine the derivatives $D_{(0,1,1)}u$ and $D_{(0,1,1)}v$ of the implicitly defined functions $u = u(x, y, z)$ and $v = v(x, y, z)$.

7.6. Spherical Coordinates. The mapping $\Phi\colon (0, \infty) \times (0, \pi) \times (-\pi, \pi) \to \mathbb{R}^3$ defined by

$$\Phi(r, \theta, \varphi) = \begin{pmatrix} r \sin\theta \cos\varphi \\ r \sin\theta \sin\varphi \\ r \cos\theta \end{pmatrix}$$

is called *spherical coordinates*.

1. Sketch the images of $r \mapsto \Phi(r, \theta_0, \varphi_0)$, $\theta \mapsto \Phi(r_0, \theta, \varphi_0)$ and $\varphi \mapsto \Phi(r_0, \theta_0, \varphi)$ for some fixed $r_0 \in (0, \infty)$, $\theta_0 \in (0, \pi)$, $\varphi_0 \in (-\pi, \pi)$.
2. What is the image of Φ?
3. Show that $\det(D_{(r,\theta,\varphi)}\Phi) = r^2 \sin\theta$ holds.
4. Conclude that the mapping Φ is a diffeomorphism onto its image.

7.7. Parametrized surfaces. In Section 12.2.2 of the lecture notes we gave several examples of important submanifolds of $\mathbb{R}^3$ (sphere $\mathbb{S}^2$, torus $\mathbb{T}^2$, Möbius strip, catenoid), in various representations.

1. Compute the tangent spaces and normal vectors to the torus, the Mobius strip, and the catenoid from their given parametrizations:
   - **Torus:**

     $$\begin{aligned} x(\phi, \theta) &= (R_1 + R_2 \cos\theta) \cos\phi, \\ y(\phi, \theta) &= (R_1 + R_2 \cos\theta) \sin\phi, \\ z(\phi, \theta) &= R_2 \sin\theta, \end{aligned}$$

     where $\phi, \theta \in [0, 2\pi)$ and $R_1 > R_2 > 0$.
   - **Möbius Strip:**

     $$\begin{aligned} x(u, v) &= \left(1 + \frac{v}{2} \cos\frac{u}{2}\right) \cos u, \\ y(u, v) &= \left(1 + \frac{v}{2} \cos\frac{u}{2}\right) \sin u, \\ z(u, v) &= \frac{v}{2} \sin\frac{u}{2}, \end{aligned}$$

     where $u \in [0, 2\pi)$ and $v \in [-1, 1]$.

- **Catenoid:**

$$\begin{aligned} x(u,v) &= \cosh(v)\cos(u), \\ y(u,v) &= \cosh(v)\sin(u), \\ z(u,v) &= v, \end{aligned}$$

where $u \in [0, 2\pi)$ and $v \in \mathbb{R}$, with cosh denoting the hyperbolic cosine function.

2. What happens to the normal vector to the Möbius strip as $u \to 2\pi$? Interpret your result geometrically.

3. Give a parametrisation for a general ellipsoid, defined as

$$\left\{(x,y,z) \in \mathbb{R}^3 : \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1\right\},$$

where $a, b, c > 0$.

7.8. Graphs as parametrized surfaces. Let $f(x,y) : D \subset \mathbb{R}^2 \to \mathbb{R}$ be a C^1 function, and let $M \subset \mathbb{R}^3$ denote its graph.

1. Define a parametrization $\phi : D \to \mathbb{R}^3$ of M explicitly.

2. Let $p \in M$, $p_3 = f(p_1, p_2)$. Compute a basis of the tangent space $T_pM \subset \mathbb{R}^3$ of M at p in terms of f.

3. Give an equation for T_pM.

4. Compute an expression for the normal vector N_pM of M at p, with positive vertical orientation, in terms of f. How does this relate to ∇f? Can you interpret this geometrically?

7.9. The IFT is only a sufficient condition. We consider the function $f(x,y) = y^2(1-x) - x^3$ from Exercise 6.3.3 in more detail.

1. Show that we cannot conclude from the implicit function theorem that f is solvable for x in a neighborhood of $(0,0)$.

2. Show, however, that the equation $f(x,y) = 0$ can be uniquely solved for x everywhere. Hint: Analyze the mapping $x \mapsto \frac{x^3}{1-x}$ on a suitable domain.

3. Denote by $Y(x) > 0$ the function such that $f(x, Y(x)) = 0$ around $x = 1/2$. Compute $Y''(1/2)$. Hint: Derive twice with respect to x the identity $f(x, Y(x)) = 0$ end evaluate it at $x = 1/2$.